1. Core AI & Machine Learning

1. Machine Learning
A field of AI where systems learn patterns directly from data rather than following hand-written rules. The model improves its performance automatically as it sees more examples. It powers everything from spam filters to recommendation engines.

2. Deep Learning
A subset of machine learning that uses multi-layered neural networks to model very complex patterns. Each layer extracts increasingly abstract features from the raw input. It is the foundation behind modern image recognition and language models.

3. Neural Network
A computing model loosely inspired by the brain, built from layers of connected nodes that transform input data. Each connection has a weight that is adjusted during training. Stacking many layers lets the network learn intricate relationships in data.

4. Supervised Learning
Training a model on labeled input-output pairs so it learns to predict outputs for new, unseen inputs. The model compares its predictions to the correct answers and adjusts itself to reduce errors. It is widely used for classification and regression tasks.

5. Unsupervised Learning
Finding hidden structure or patterns in unlabeled data without being told the correct answers in advance. Common techniques include clustering similar items together and reducing data to fewer dimensions. It is useful for exploring data when labels are unavailable.

6. Reinforcement Learning
Training an agent to make a sequence of decisions by rewarding desired behavior and penalizing undesired behavior. The agent learns through trial and error while interacting with an environment. It has been used to master games and control robotic systems.

7. Overfitting
A problem where a model learns the training data too closely, including its noise and quirks, and fails to generalize. It performs very well on data it has already seen but poorly on new data. Techniques like regularization and more data help reduce it.

8. Gradient Descent
An optimization algorithm that iteratively adjusts a model's parameters to minimize its error on the training data. It works by moving parameters in the direction that most reduces the loss function. Variants like stochastic gradient descent make it practical for huge datasets.

9. Backpropagation
The algorithm used to calculate how much each weight in a neural network contributed to the final error. It propagates the error backward through the network, layer by layer. This allows gradient descent to update every weight efficiently during training.

10. Feature Engineering

The process of selecting, transforming, and creating input variables that help a model learn more effectively. Good features can make a simple model outperform a complex one with poor inputs. It often requires domain knowledge about the problem being solved.

2. Generative AI & Large Language Models

11. Large Language Model (LLM)
A neural network trained on massive amounts of text so it can understand and generate human-like language. It learns statistical patterns of how words and ideas relate to one another. Modern LLMs can write, summarize, translate, and answer questions across many domains.

12. Transformer Architecture
A neural network design built around self-attention that underlies most modern language and vision models. Unlike older architectures, it processes entire sequences in parallel rather than one step at a time. This made it possible to train much larger and more capable models.

13. Attention Mechanism
A technique that lets a model weigh how relevant different parts of the input are when producing each part of the output. It allows the model to focus on the most important words or tokens, even if they are far apart. This is a core building block of transformer models.

14. Tokenization
The process of breaking text into smaller units, called tokens, that a language model can process numerically. Tokens can be whole words, parts of words, or even individual characters. How text is tokenized affects both model performance and cost.

15. Prompt Engineering
The practice of carefully crafting inputs to guide an AI model toward better, more specific, or more reliable outputs. It involves choosing wording, structure, and examples that steer the model's behavior. Effective prompts can dramatically improve results without changing the model itself.

16. Fine-Tuning
The process of further training a pretrained model on a smaller, specialized dataset to adapt it to a specific task. It builds on the general knowledge the model already has rather than starting from scratch. This makes it much cheaper than training a new model from the ground up.

17. Retrieval-Augmented Generation (RAG)
A method where a model retrieves relevant external documents before generating its response, grounding it in real information. This reduces reliance on the model's internal memory alone, which can be outdated or incomplete. It is commonly used to build chatbots that answer questions over private documents.

18. Hallucination
A failure mode where an AI model generates confident-sounding but factually incorrect or entirely fabricated information. It happens because the model predicts plausible-sounding text rather than verifying facts. Careful prompting, grounding techniques like RAG, and human review help mitigate it.

19. Diffusion Model
A generative model that creates images, audio, or other data by starting from random noise and gradually refining it. During training, it learns to reverse a process that slowly adds noise to real data. This step-by-step denoising approach powers many popular image-generation tools.

20. Multimodal AI

AI systems that can understand and generate multiple types of data together, such as text, images, audio, and video. This lets a single model answer questions about a picture or generate captions for a video. It moves AI closer to how humans naturally combine different senses.

21. Context Window

The maximum amount of text, measured in tokens, that a language model can consider at one time when generating a response. Anything beyond this window is effectively invisible to the model during that interaction. Larger context windows allow models to work with longer documents and conversations.

22. Embedding

A numerical vector representation of data, such as words, sentences, or images, that captures its underlying meaning. Items with similar meaning end up close together in this vector space. Embeddings power search, recommendation systems, and retrieval-augmented generation.

23. Agentic AI

AI systems designed to autonomously plan, decide, and execute multi-step tasks with limited ongoing human input. An agent might break a goal into subtasks, use tools, and adjust its plan based on results. This is an emerging direction beyond simple question-answering AI.

24. AI Alignment

The ongoing effort to ensure AI systems act in accordance with human values, intentions, and safety expectations. It involves designing training methods, guardrails, and evaluation techniques that catch harmful or unintended behavior. Alignment becomes more important as AI systems gain more autonomy.

3. Data & Infrastructure

25. Big Data
Extremely large or complex datasets that traditional tools struggle to store, process, and analyze efficiently. It is often described using the three Vs: volume, velocity, and variety of data. Specialized frameworks and distributed systems are needed to make sense of it.

26. Data Lake
A centralized repository that stores large amounts of raw data in its native format, structured or unstructured. Unlike a traditional database, it does not require data to be organized before storage. This flexibility makes it useful for exploratory analysis and machine learning.

27. Data Pipeline
An automated sequence of steps that moves and transforms data from its source to its final destination. It typically handles collection, cleaning, transformation, and loading in a repeatable way. Reliable pipelines are essential for keeping analytics and AI models fed with fresh data.

28. ETL (Extract, Transform, Load)
A process for extracting data from various sources, transforming it into a usable format, and loading it into a target system. It is a foundational pattern in data engineering and business intelligence. Modern variants like ELT shift the transformation step to happen after loading.

29. Vector Database
A database specifically optimized for storing and searching high-dimensional embedding vectors by similarity. It enables fast lookups like finding documents with meaning similar to a query. Vector databases are a key component behind retrieval-augmented generation systems.

30. GPU Computing
Using graphics processing units, originally designed for rendering images, to accelerate parallel computations. Their architecture is well suited to the matrix operations that underpin neural network training. GPUs have become essential hardware for both training and running AI models.

31. Distributed Computing
An approach that splits computational tasks across multiple machines working together to increase speed and scale. It allows systems to process far more data than a single computer could handle alone. Coordinating these machines reliably is a major challenge in system design.

32. Data Governance
The policies, processes, and standards that ensure data across an organization is accurate, secure, and used responsibly. It covers data quality, access control, privacy compliance, and accountability. Strong governance builds trust in the data that AI and analytics rely on.

4. Cloud, DevOps & Systems

33. Cloud Computing
A model for delivering computing resources like storage, servers, and processing power over the internet on demand. Users pay for what they use rather than owning physical hardware outright. This flexibility has made it the backbone of most modern software infrastructure.

34. Containerization
The practice of packaging software together with all its dependencies into isolated, portable units called containers. This ensures an application runs the same way regardless of where it is deployed. Docker is the most widely used tool for building and running containers.

35. Kubernetes
An open-source system for automating the deployment, scaling, and management of containerized applications. It handles tasks like restarting failed containers and distributing load across machines. It has become the standard for running large-scale, containerized systems in production.

36. Microservices
An architectural style where an application is built as a set of small, independently deployable services. Each service handles a specific piece of functionality and communicates with others over the network. This makes large systems easier to develop, scale, and update independently.

37. API (Application Programming Interface)
A defined set of rules and protocols that lets different software systems communicate with one another. It specifies what requests can be made, how to make them, and what responses to expect. APIs are the connective tissue behind nearly all modern software integrations.

38. CI/CD
Continuous Integration and Continuous Delivery, a set of practices that automate testing and deployment of code changes. Continuous integration merges and tests code changes frequently to catch issues early. Continuous delivery then automates releasing that code safely and reliably to production.

39. Serverless Computing
A cloud model where the provider automatically manages server infrastructure, scaling resources up and down as needed. Developers write code without worrying about provisioning or maintaining servers directly. It can reduce operational overhead and cost for workloads with variable demand.

40. Edge Computing
An approach that processes data closer to where it is generated, such as on local devices, rather than in a distant data center. This reduces latency and bandwidth use, which matters for real-time applications. It is increasingly important for IoT devices and autonomous systems.

5. Security & Emerging Tech

41. Zero Trust Security
A security model that assumes no user or device should be trusted by default, even inside the network perimeter. Every request must be continuously verified based on identity, context, and device health. This approach reduces the damage a single compromised account or device can cause.

42. Blockchain
A distributed, tamper-resistant ledger technology that records transactions across many computers simultaneously. Once data is added, it is extremely difficult to alter without detection, thanks to cryptographic linking. It underpins cryptocurrencies but is also explored for supply chains and record-keeping.

43. Quantum Computing
A computing paradigm that uses quantum bits, or qubits, to perform certain calculations far faster than classical computers. Qubits can represent multiple states simultaneously, enabling new kinds of parallelism. It is still an emerging field, but it holds promise for cryptography, chemistry, and optimization.

44. Digital Twin
A virtual replica of a physical object, process, or system, used for simulation, monitoring, and prediction. Sensors on the real object feed live data into the digital model to keep it synchronized. Industries use digital twins to test changes safely before applying them in the real world.

45. Internet of Things (IoT)
A network of physical devices embedded with sensors and software that collect and exchange data over the internet. These devices range from smart home gadgets to industrial equipment monitoring systems. IoT enables automation and real-time insight into physical environments.

46. Federated Learning
A technique for training machine learning models across many decentralized devices without moving raw data to a central server. Each device trains on its own local data and shares only model updates, not the data itself. This helps protect user privacy while still benefiting from a shared, improved model.

47. Explainable AI (XAI)
A set of techniques that make AI model decisions more interpretable and understandable to humans. It helps answer why a model produced a particular prediction or recommendation. This is especially important in regulated fields like healthcare and finance.

48. Synthetic Data
Artificially generated data that mimics the statistical properties of real data, used when real data is scarce or sensitive. It can help train models while avoiding privacy risks tied to real personal information. Synthetic data is increasingly used to fill gaps and balance underrepresented cases.

49. Model Compression
A set of techniques, including pruning and quantization, that shrink AI models so they run faster on limited hardware. Pruning removes unnecessary parts of a network, while quantization reduces the precision of its numbers. This makes it possible to run capable models on phones and edge devices.

50. AI Governance

The frameworks, policies, and regulations designed to guide the responsible development and deployment of AI systems. It addresses concerns like fairness, transparency, accountability, and safety. As AI becomes more powerful, governance aims to balance innovation with protecting society.